Bach Nguyen

602-565-5951 | LinkedIn | GitHub | bachnd05@gmail.com

Education

Arizona State University Tempe, AZ
Bachelor of Science in Computer Science *08/2023 – Expected 05/2027*
- GPA: 4.0/4.0, Specialty: NLP, Computer Vision and Reasoning in Multimodal Models

Work Experience

Research Assistant in AI Reasoning Lab 05/2025 – Present
Arizona State University *Tempe, AZ*
- Collaborated with Professor Ben Zhou in ARC Lab and PhD researchers on multimodal LLM reasoning and decision-making under uncertainty, contributing to novel frameworks for handling model hallucinations
- Led a research on active reasoning benchmark for multimodal LLMs enabling iterative uncertainty resolution
- Co-authored benchmark paper evaluating vision-language models' reasoning capabilities and performance on fundamental vision analogies including object detection, spatial reasoning, and visual understanding

Mathematics Teaching Assistant 08/2024 – Present
Arizona State University *Tempe, AZ*
- Assisted in teaching College Algebra, Discrete Math, and Calculus II in classes of around 80 students by providing mentoring, grading problem sets and assignments
- Designed and facilitated adaptive group study sessions and instructional materials tailored to individual student learning needs, improving overall class performance and engagement

Publications

"How Well Do LLMs Reason with Noisy Evidence? An Active Visual Reasoning Benchmark."
First Author *Under Review at NAACL 2027.*

"VisAnalog: A Diagnostic Suite for Visual Concept Transfer on Natural Images."
Co-Author (in collaboration with UC Davis) *Accepted at CVPR Workshop VisCon 2026.*

Projects

Active Reasoning - Noisy Agentic Interaction | *vLLM, PyTorch* 06/2025 – 3/2026
- Architected active reasoning system where text-only LLMs orchestrate strategic multi-turn dialogues with VLMs to reduce hallucinations in visual question answering, implementing probabilistic belief updates that achieve 85%+ confidence thresholds before final answer generation
- Implemented adaptive questioning strategy with Bayesian belief state tracking that dynamically generates targeted follow-up queries based on response consistency and ensemble sampling (temperature=1.0), achieving 20% accuracy improvement over traditional end-to-end VQA pipelines

DeepSeek-R1 Fine-Tune and Benchmark | *Transformers, Unsloth* 02/2025 – 02/42025
- Fine-tuned DeepSeek-R1-7B model on medical reasoning dataset using QLoRA (4-bit quantization + LoRA), achieving statistically significant accuracy improvement on BioASQ benchmark validated through 10-fold cross-validation
- Optimized training pipeline on GPU-constrained environment through parameter-efficient fine-tuning techniques, reducing memory footprint by 35% while maintaining model performance

OpenCV Object Detection and Image Processing | *OpenCV, Numpy* 03/2024 – 06/2024
- Developed real-time object detection system using OpenCV and Haar Cascade classifiers to accurately identify faces and vehicles with 90%+ detection accuracy
- Conducted extensive testing and optimization across various lighting conditions and environments, implementing optimized convolution kernels for edge detection to improve processing speed by 30%

Technical Skills

Languages: Python, Java, C/C++, MATLAB, HTML, JavaScript, CSS, SQL
Frameworks and Libraries: PyTorch, vLLM, Transformers, Gradio, Unsloth, OpenCV, Flask, Pandas, Git/GitHub
Tools and Platforms: AWS, CUDA, Linux/Ubuntu, GPU Computing (A100/H200), Docker, tmux, GDB
Relevant Knowledge: RAG, RLHF, Computer Vision, Statistical Analysis, CNNs, RNNs, Transformers, BiLSTM